The Effect of Paid Search Advertising on Revenue: A Replication of the eBay Natural Experiment

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1 Introduction

Firms spend billions of dollars each year on search engine marketing (SEM), yet measuring its causal effect on sales is difficult. When a consumer searches for a product online, it is often unclear whether a paid advertisement actually drives the purchase or whether the consumer would have found the company through organic search results anyway. This makes it challenging for firms to determine the true return on investment of paid search advertising.

This paper examines the effect of paid search advertising on eBay's revenue using a natural experiment studied by Blake, Nosko, and Tadelis (2014). In this experiment, eBay temporarily stopped bidding on search advertisements in a subset of geographic markets while continuing advertising in others. By comparing changes in revenue between these groups before and after the change, we can estimate the causal impact of paid search using a difference-in-differences (DID) approach.

2 Experimental Setting

The analysis is based on a large-scale field experiment conducted by eBay and studied by Blake, Nosko, and Tadelis (2014). Beginning on May 22, 2012, eBay stopped bidding on Google AdWords advertisements in 65 designated market areas (DMAs) in the United States. These markets form the treatment group. In the remaining 145 DMAs, eBay continued its paid search advertising as usual, forming the control group. The experiment lasted for approximately eight weeks.

Importantly, the experiment was not fully randomized. The largest DMAs were excluded from treatment because eBay expected that turning off advertising in these markets could significantly affect revenue. As a result, the treatment and control groups differ in their baseline levels of revenue, with control markets generally having higher average sales. Because of this baseline difference, a simple comparison of average revenue across groups would not provide a valid estimate of the treatment effect. Instead, this paper uses a difference-in-differences approach that compares how revenue changes over time within each group.

3 Data and Figures

The dataset contains daily revenue observations for 210 designated market areas between April and July 2012. Each observation records the date, the DMA identifier, whether the market belongs to the treatment or control group, and the revenue generated by eBay in that market on that day. The treatment period begins on May 22, 2012, when paid search advertising was turned off in the treatment DMAs.

Figure 1 shows the average daily revenue for the treatment and control groups over time. The dashed vertical line marks the start of the treatment period.

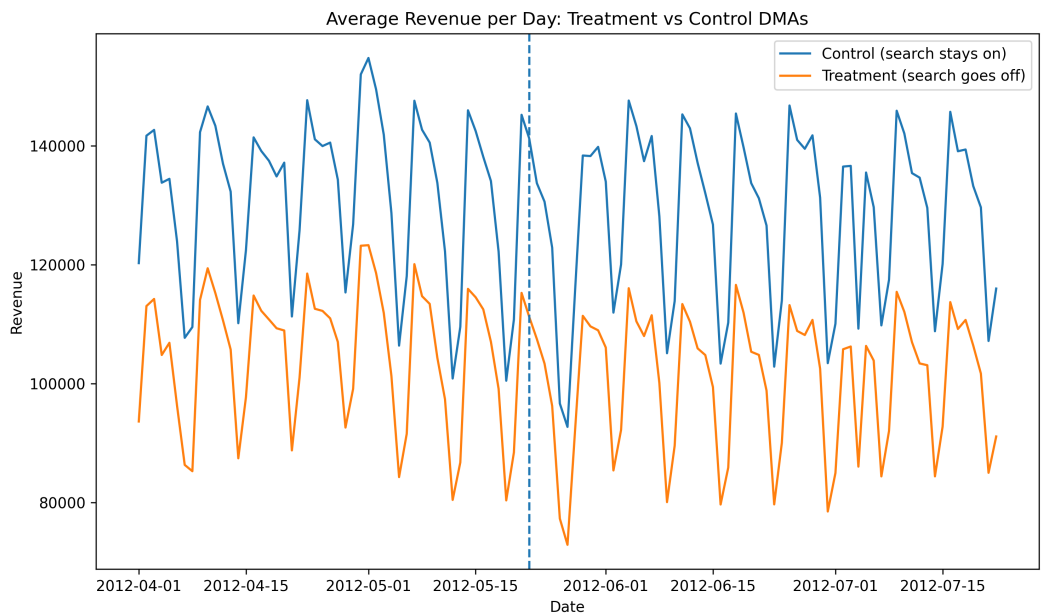

Figure 1: Average revenue for treatment and control DMAs over time. The dashed line marks May 22, 2012, when SEM was turned off for the treatment group.

The figure shows that the control group consistently has higher average revenue than the treatment group. This difference reflects the experimental design, since the largest markets were excluded from the treatment group. Both groups display similar weekly oscillation patterns, suggesting that they follow similar seasonal demand patterns. At this scale, it is difficult to visually identify a clear divergence in revenue trends after the treatment begins.

Figure 2 plots the difference in log average revenue between the control and treatment groups over time. Before May 22, the gap fluctuates around a relatively stable level, indicating that the two groups follow similar trends prior to the intervention. After the treatment begins, the gap appears to increase slightly, suggesting that revenue in the treatment markets may have fallen relative to the control markets. This visual pattern motivates the formal difference-in-differences analysis presented in the next section.

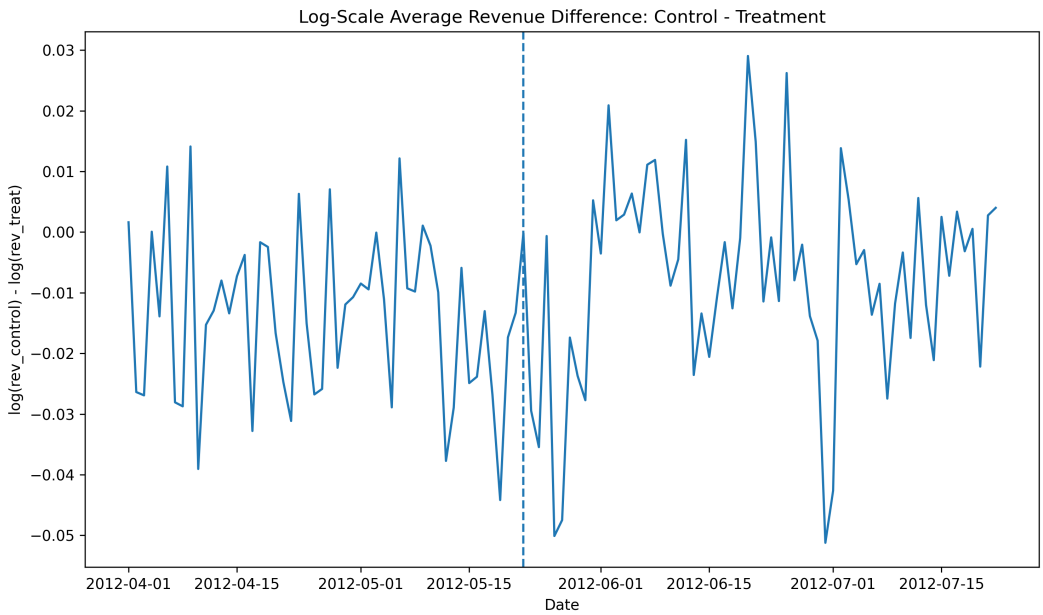

Figure 2: Log-scale average revenue difference between control and treatment groups over time. The gap appears to increase after May 22.

Table 1: Difference-in-Differences Estimate of the Effect of Paid Search on Revenue

	Log Scale	Levels (exp)
Point Estimate ($\hat{\gamma}$)	-0.0066	0.9934
Standard Error	0.0056	—
95% CI	$[-0.0175,\ 0.0043]$	$[0.9827,\ 1.0043]$

4 Results

Table 1 reports the difference-in-differences estimate of the effect of turning off paid search advertising on revenue. The estimated treatment effect is $\hat{\gamma} = -0.0066$ in log scale. Converting this estimate to levels implies that revenue in the treatment markets declined by approximately 0.66% after paid search advertising was turned off.

However, the estimated effect is not statistically significant at conventional levels. The 95% confidence interval ranges from -0.0175 to 0.0043, which includes zero. This means we cannot reject the hypothesis that paid search advertising had no measurable effect on revenue during the experiment period.

Economically, this result suggests that paid search advertising contributes little incremental revenue for a well-known brand such as eBay. Many users who search for eBay may already intend to visit the website and would find it through organic search results even in the absence of paid advertisements.

5 Conclusion

This paper replicates a difference-in-differences analysis of eBay's paid search advertising experiment. The results suggest that turning off paid search advertising reduced revenue by approximately 0.66 percent, but the effect is not statistically significant. These findings replicate the main conclusion of Blake, Nosko, and Tadelis (2014), which is that paid search advertising has a very small effect on revenue for a well-known brand.

Several limitations should be noted. The dataset used in this replication is scaled and does not represent actual revenue values, and the experiment was not fully randomized across markets. In addition, the analysis uses a simple difference-in-differences estimator rather than a more detailed regression framework with additional controls. Despite these limitations, the results highlight the importance of using experimental or quasi-experimental methods when evaluating the effectiveness of marketing strategies.

References

[1] Blake, T., C. Nosko, and S. Tadelis (2014). "Consumer Heterogeneity and Paid Search Effectiveness: A Large-Scale Field Experiment." *Econometrica*, 83(1): 155–174.

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